OCR design choices: complementary mechanisms

Examples illustrate overlapping design choices, not exclusive eras or a performance ranking.

Detection and recognition pipelines

Locate text and recognize selected regions.
Expose intermediate boxes and transcriptions.
Segmentation or recognition errors can propagate.

OCRopus; Tesseract
PP-OCR also uses neural sequence models.

Neural sequence recognition

Learn image-to-sequence mappings.
CTC or autoregressive decoding predicts text.
Output and supervision depend on the task.

CRNN; Calamari; PP-OCR
Can be used inside a modular pipeline.

Joint text and layout generation

Generate text with locations or formatted content.
Preserve structure in the output representation.
Serialization alone does not establish correctness.

VISTA-OCR; GOT-OCR
Text and structure require separate evaluation.

Vision-language OCR and parsing

Condition generation on visual tokens.
Resolution and compression affect retained detail.
External OCR can still be used during training.

Ocean-OCR; PaddleOCR-VL
TextHawk2; DeepSeek-OCR; Qwen3-VL